Hale, David K. 2026. "TrafficCube spatiotemporal analytics engine: early vision and direction." https://github.com/dkhale/TrafficCube. Accessed October 2026.

Abstract

A central axiom of modern computer science is that artificial intelligence (AI) delivers its highest value when processing large volumes of well-structured data. Despite this history, many traffic analysts and researchers are trying to leverage AI chatbots (e.g., ChatGPT, Claude) to expedite their analyses. Although capable of impressive feats, chatbots may be prone to oversights and hallucinations when forced to operate on ad-hoc datasets. To fill this gap, a new data analytics platform called TrafficCube seeks to bring AI-powered traffic analysis to the general public. TrafficCube is available free of charge and uses a simple file format that anyone can work with. This technical paper will describe and demonstrate TrafficCube's approach to fusing, visualizing, and simultaneously analyzing multiple datasets from multiple sources. This technical paper will also discuss available and upcoming opportunities to perform AI-assisted similarity searches, contributing factor analyses, simulation calibrations and validations, data quality and anomaly identifications, Highway Capacity Manual methods, short-term predictions, pattern recognitions, congestion diagnoses, and Traffic Analysis Toolbox Volume 3 analyses, among other opportunities.

KEYWORDS:

Data analytics, artificial intelligence, traffic analysis

Motivation

In recent decades, traffic engineering consultants and State-local transportation agency staff have rarely brought strong backgrounds in data science or industrial engineering, although this educational and training gap may be slowly improving. Instead, traffic analyses have often centered around traditional modeling approaches such as highway capacity analysis and microsimulation. More recently, leading traffic researchers and developers have encouraged practitioners to increase their use of data analytics tools and methods, to complement and enhance those traditional approaches. This is due to rapidly evolving data sources, data collection technologies, data management platforms, and the analytical methods themselves, especially those involving artificial intelligence (AI). As such, practitioners and agency staff are doing their best to learn and adopt the new tools and methods, with varying degrees of success. Some examples of these data analytics tools and methods include:

- PTV Flows and Optima (1)
- Caltrans Traffic Mobility Insights (2)
- Texas Department of Transportation DAVE (2)
- FHWA Traffic Analysis Toolbox Volume 3 (2019 edition) (3)
- Regional Integrated Transportation Information System (RITIS) (4)

Despite the availability of new tools and methods, many private consultants and public agency staff are not highly familiar with these new methods, and are hesitant to learn and apply them, for various reasons. For example, proprietary products (e.g., Flows, Optima) often do not disclose their computational methods, require significant license fees, and are not always compatible with available data sources and formats. Online platforms (e.g., DAVE) are not available to all practitioners and researchers and may lack certain valuable analysis methods. Data-driven computational methods (e.g., TAT Volume 3) are not well understood by many and often lack efficient software automation. Therefore, there continues to be a need for one or more lightweight data analytics tools that are easy-to-use, ingest a wider variety of data formats, and encourage adoption of statistical and AI-based analysis methods.

High-Level Solution Approach

TrafficCube (see figure 1) is a spatiotemporal traffic analysis platform that allows AI to operate on large quantities of well-structured data (e.g., months or years-worth of traffic volumes, probe data, weather data, crash data, vehicle trajectories, traffic analysis tool input and output data). The tool fills a key niche in the industry by providing:

- The simplest possible data exchange format
- Efficient visualization to facilitate human auditing of ingested data
- Embedded intelligence to recognize and classify common traffic data
- Automated methods to normalize ingested data and perform statistical analyses
- Versatile features to let users define and analyze edge cases that are less common
- Advanced memory management to prevent performance issues with huge datasets

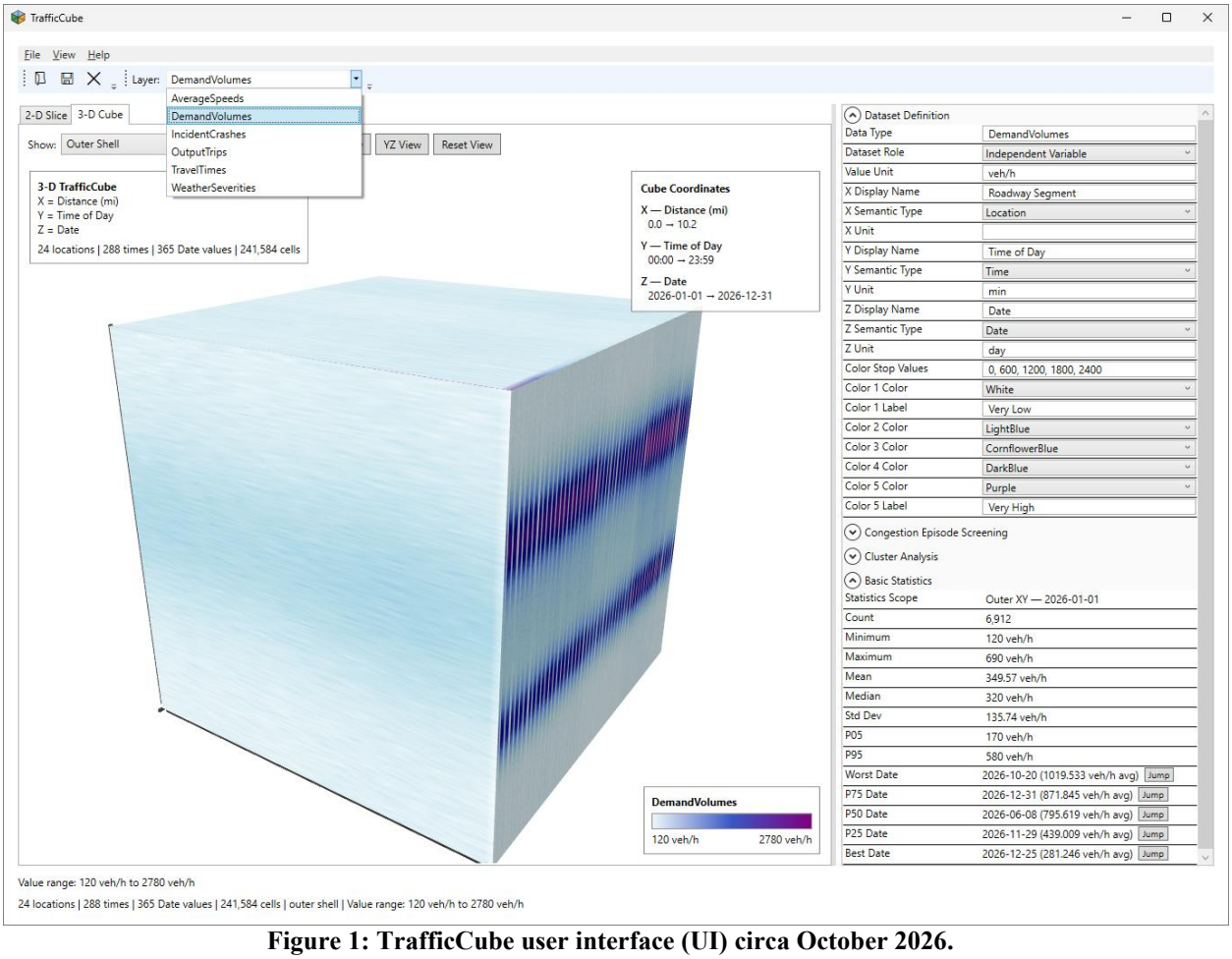


Figure 1: TrafficCube user interface (UI) circa October 2026.

At the AI Applications in Urban Transportation Management Symposium at New York University (September 2026), one speaker said, "All transportation data have a time and a location." This exact sentiment led to initial versions of TrafficCube assuming location data on the X-axis, times-of-day on the Y-axis, and days-of-year or random number seeds on the Z-axis. Newer versions let users override those defaults (if desired) and define custom data types (see Dataset Definition in figure 1). And although the tool contains intelligence to estimate and assign thresholds and settings for common traffic data types, users can also override those thresholds and settings if they wish. For example, if TrafficCube believes it is ingesting travel times, it will assign units of measurement (i.e., seconds if the travel times are high, minutes if the travel times are low), five logical travel time ranges, and five typical colors (i.e., ranging from green to red) for representing those logical ranges, all of which users can customize under Dataset Definition if they prefer. In the figure 1 demonstration example, the user has ingested and fused six data types:

- Average speeds (dependent variable)
- Demand volumes (independent variable)
- Incident crashes (independent variable)
- Output trips (dependent variable)
- Travel times (dependent variable)
- Weather severities (independent variable)

Now that the user has assembled their multi-source data into a well-structured schema, the tool has what it needs to perform advanced statistical or AI-based (e.g., via open-source machine learning libraries, or agentic AI users) analyses such as:

- Similarity searches
- Agentic AI applications
- Contributing factor analysis
- Before-and-after assessment
- Simulation calibration and validation
- Data quality and anomaly identification
- Annual travel time reliability visualizations
- Machine learning and short-term prediction
- Highway Capacity Manual (HCM) methods
- Pattern recognition and congestion diagnosis
- AI-based traffic analysis insights and recommendations
- Traffic Analysis Toolbox (TAT) Volume 3 full automation

Looking forwards, the logical roadmap for functionality development consists of 1) multi-source traffic data fusion and visualization, 2) basic statistical analyses and visualizations not requiring AI, 3) advanced statistical analyses and visualizations not requiring AI, 4) lightweight AI analyses and pattern recognitions, 5) advanced AI analyses involving traffic state predictions while trained and informed by a corpus of industry literature. TrafficCube is 2.5 months old at the time of this writing and is currently within steps 2 and 3, continuously assimilating new statistical analyses and visualizations not requiring AI. However, the strong foundation of visualization, statistics, and management of large datasets will uniquely position TrafficCube to perform the AI analyses envisioned in steps 4 and 5 of the roadmap.

Software Engineering Approach

TrafficCube development is being accelerated by AI-enabled computer programming. Figure 2 illustrates the integrated development workflow that features OpenAI Codex for code writing, GitHub for code archiving, and Microsoft Visual Studio for tool building). After one month of development (August 2026), the tool could fuse and visualize multi-source data in 2D or 3D mode. After the second month of development (September 2026), the tool could compute basic statistics (e.g., mean, median, standard deviation, 95th percentile), screen for congestion episodes exceeding user-defined thresholds, perform cluster analysis to identify similar days (e.g., low demand, high demand, high demand plus incident, incident plus severe weather, high demand plus severe weather), and representative day analysis to statistically identify which days best represent their cluster. Priorities for month #3 (October 2026) include microsimulation calibration support, plus congestion (or incident) causal analysis, diagnosis, and visualization (e.g., congestion causal pie charts).

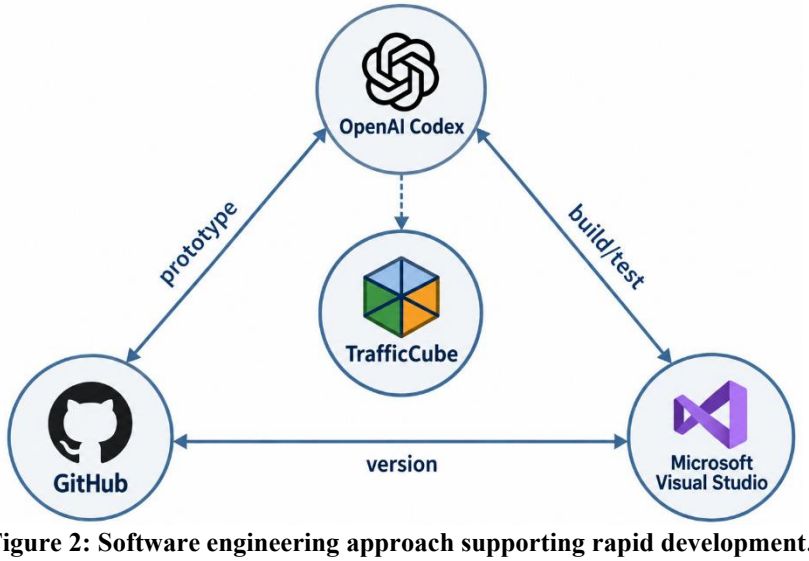


Figure 2: Software engineering approach supporting rapid development.

The software architecture seeks to preserve flexibility as the concept matures. The current desktop implementation uses Windows Presentation Foundation (WPF) within the Visual Studio and .NET development environment because that stack supports rapid prototyping of interactive graphics, file ingestion, and local analytical workflows on ordinary Windows computers. At the same time, the calculation and data management logic were designed to be separable from the user interface, so that importing data, defining cube dimensions, normalizing measures, screening episodes, clustering conditions, and producing analytical summaries will not permanently tie to a specific desktop display. This separation creates a path for future implementations in which the analytical engine could support different front ends, including a WPF desktop application, a Blazor web interface, or other agency and research tools that call the same core logic. In practical terms, the architecture treats TrafficCube as a reusable analytical engine with an evolving set of interfaces, allowing early development to proceed quickly on the desktop while preserving the possibility of future web deployment, cloud-assisted processing, agentic AI processing, or integration into larger transportation data systems.

TrafficCube's memory-management logic is designed to keep large multi-dimensional datasets responsive during loading, layer switching, and 3D interaction. Instead of treating every visual cell as a fully independent object, the program can store shared axis definitions once, store measure values in compact indexed arrays or equivalent structures, and generate visual geometry from those data structures only as needed. For 3D rendering, the software does not need to draw every underlying observation at full detail during every rotation; it can use decimation, cached geometry, selected Z slices, or simplified visual surfaces so that user interaction remains smooth. When the user switches layers, the coordinate grid can remain stable while only the active measure values, color scale, legend, and threshold-dependent highlights are refreshed. Similarly, when the cube is rotated, the underlying analytical data do not need to be recalculated; the rendering system only updates the view transformation and redraws the current visual representation. This distinction between analytical storage and visual rendering is important to TrafficCube's practical niche: the tool can preserve detailed data for analysis while using lighter-weight visual objects for interaction, reducing the risk that realistic traffic datasets will overwhelm memory or make the interface sluggish.

TrafficCube development began on August 1, 2026. The first public release of TrafficCube (on GitHub) was in mid-September 2026. Since then, at the time of this writing (i.e., mid-October 2026), upgrade releases have been posted on a weekly basis. New releases will be posted on a weekly or monthly basis for the foreseeable future. All releases to this point have essentially constituted a soft launch (i.e., word-of-mouth) because most of the advanced statistical and AI analysis methods are not implemented yet. However, the plan is to gradually announce the tool more broadly as it continues to strengthen, and as more computational analysis methods are completed.

Data Fusion Approach

Different organizations have tried to promote universal traffic input-output (I/O) data formats (e.g., 1990s Traffic Model Markup Language, 2000s Universal Traffic Data Format, 2010s General Modeling Network Specification) (5, 6, 7). However, their complexity may have prevented (and may currently be preventing) industry-wide adoption, so a simpler format may be needed. The FHWA Congestion and Bottleneck Identification (CBI) tool (8) and HCM travel time reliability procedures (9) adopted a simpler schema, a three-dimensional (3D) analysis cube. Figure 3 shows versions of this cube having road segments on the X-axis, time-of-day on the Y-axis, and day-of-year on the Z-axis. This simple schema can unify many relevant data types (e.g., input free-flow speeds, traffic volume demands, output travel speeds and travel times, traffic counts and other detector data, weather and incident data, probe data). Although TrafficCube uses this schema by default, users can customize the data types and meanings on each axis as discussed in the High-Level Solution Approach section.

TrafficCube imports simple comma-separated value (CSV) files that specify data values for the X and Y axes, and for the Z-axis if needed. Figure 4 illustrates the simple format. TrafficCube estimates the information shown in the Dataset Definition section of the properties panel by scanning the ingested records before visualization. During this pass, the program identifies the distinct values on each coordinate axis, counts the resulting X, Y, and Z dimensions, detects whether the Y-axis appears to use a regular time interval, and derives the first and last time points, value range, missing-coordinate patterns, and simple descriptive statistics. When users supply optional metadata under Dataset Definition (e.g., axis labels, measure names, units, color stops, threshold definitions), TrafficCube can use those values directly. When metadata are absent, the tool infers likely characteristics from the data itself. For example, a regularly spaced sequence of time-of-day values can be treated as the observation interval, a date-like Z value can be labeled as a date axis, and a numeric Z value with repeated runs can be interpreted as a scenario or seed axis if the surrounding metadata support that interpretation. The resulting logic is intentionally conservative: values that can be calculated directly are reported as facts, values inferred from record patterns are treated as provisional metadata, and uncertain assumptions are disclosed to the analyst rather than hidden inside the import process.

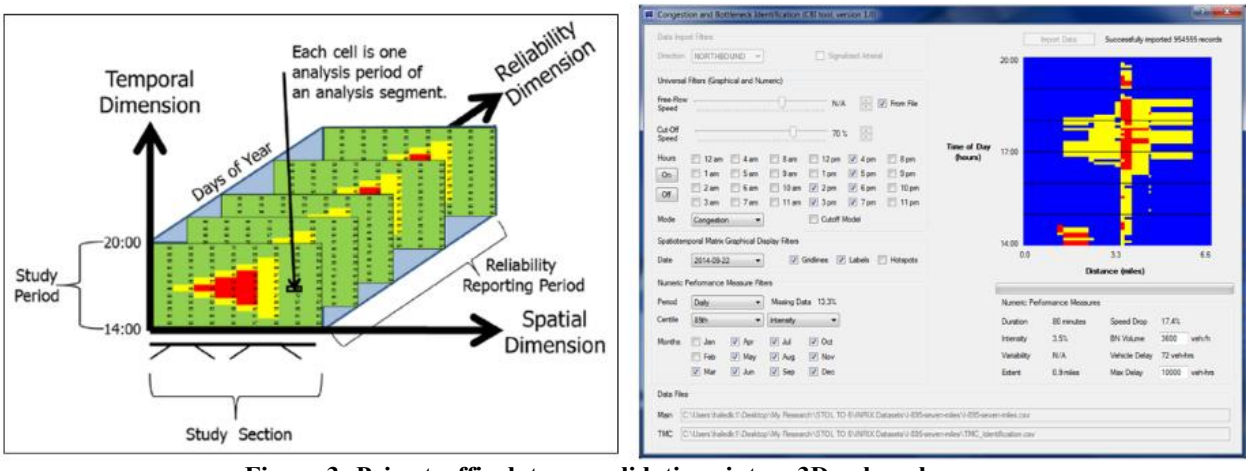


Figure 3: Prior traffic data consolidations into a 3D cube schema.

	X	Y	Z	
Loc ID	Dist	Time	Date	Value
11+01	1.49	6:00	0121	65.5
11+02	1.07	6:00	0121	65.0
11+03	0.53	6:00	0121	65.7
11+04	0.80	6:00	0121	65.2
11+05	1.14	6:00	0121	65.7

Figure 4: TrafficCube comma-separated value (CSV) file format.

TrafficCube normalizes ingested data by converting records from a source-table form into a consistent cube-index form. The import process builds lookup tables for each axis: every unique location becomes an X index, every unique time interval becomes a Y index, and every unique date, seed, scenario, or other third-axis value becomes a Z index. The program then maps each incoming record from its original coordinate labels into the corresponding cube coordinate and stores the measure value at that indexed position. This approach allows different data types (e.g., speed, volume, weather) to share the same cube dimensions even when they arrive in separate files. If a measure is missing a coordinate that exists in another measure, the cube can preserve the shared grid while marking that value as missing rather than distorting the dimensional structure. If a file contains extra coordinates or incompatible axis definitions, the import process can detect the mismatch and warn the analyst. The key algorithmic idea is that TrafficCube separates coordinate normalization from measure storage: once the axes are reconciled, multiple measures can be compared, sliced, clustered, or screened for episodes using the same spatial and temporal frame.

Sample Analysis Package

As alluded to in the figure 1 demonstration example, the online release provides six demonstration datasets to help new users understand TrafficCube as both a visualization environment and an analytical framework. These are lightweight datasets (i.e., approximately 2 MB apiece) having only two weeks and 24 segments of observed data for average speeds, demand volumes, incident crashes, output trips, travel times, and weather severities. These layers show how independent variables such as weather can compound to affect dependent traffic performance measures such as travel time. For example, higher demand volumes may produce only moderate delay under normal conditions, but the same demands can generate sharper speed reductions and travel time increases when combined with adverse weather or incident conditions (see figure 5). Because the cube preserves the location, time-of-day, and date dimensions, users can also see how congestion is not isolated to a single cell: a bottleneck can create lower speeds on upstream segments, and the effects can persist into subsequent time intervals even after the initial contributing condition begins to subside. The demo datasets therefore serve as a controlled learning environment, allowing users to observe how multi-dimensional traffic patterns appear visually in 2D and 3D views while also producing measurable statistical signatures such as changed value ranges, cluster assignments, representative days, and congestion episode outputs.

Although the lightweight datasets embedded in the software download contain only two weeks of data, the GitHub site offers an optional download for full-year versions of these same datasets (i.e., a downloadable ZIP file of approximately 48 MB, unzipped CSV files of approximately 448 MB). Due to OpenAI-Codex's inclusion of advanced data and memory management in TrafficCube, the full-year traffic datasets show that users can manipulate the 2D and 3D views without the usual software performance delays associated with large datasets. The full-year datasets also demonstrate typical annual traffic behaviors and patterns (e.g., light traffic on holidays, traffic volume surges in the early Fall, increased frequency of severe weather in the evenings, medium traffic demands during mid-day off-peak periods, increased frequency of crashes during peak commuting times).

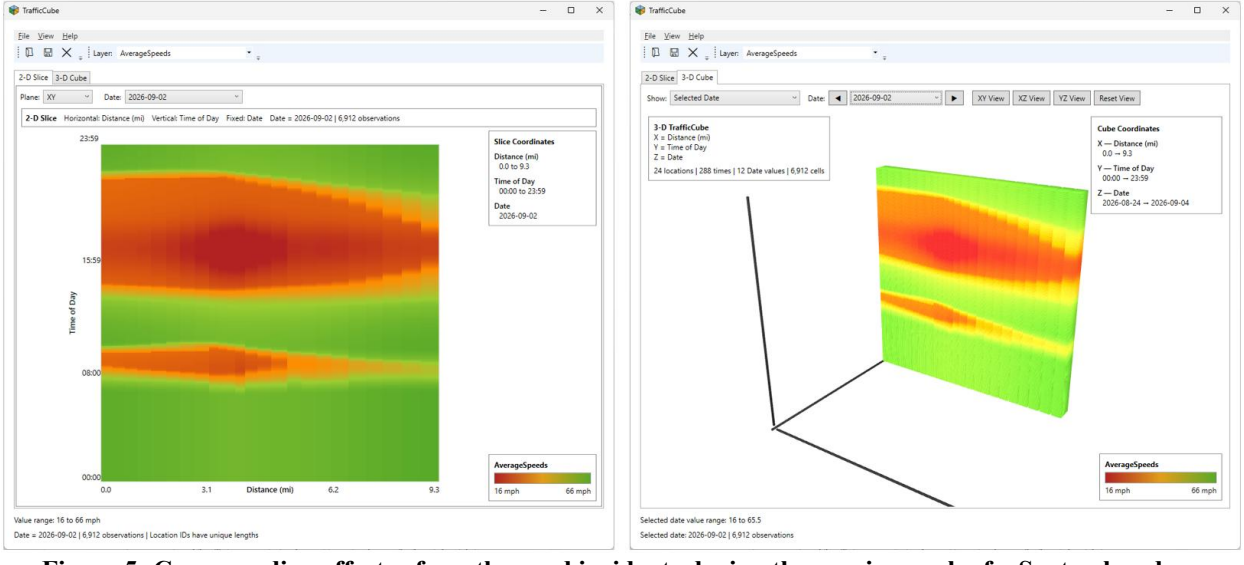


Figure 5: Compounding effects of weather and incidents during the evening peak of a September day.

Calibration Automation: An Early Demonstration of Unique Capability

TrafficCube can quite naturally assist verification, calibration, and validation (VC&V) efforts by comparing traffic simulation input (e.g., demand volumes) and output (e.g., average speeds) layers to field-measured dependent (e.g., output trips) and independent variable (e.g., weather severities) layers. It is common these days for practitioners to supplement their calibration and validation efforts by comparing simulation outputs to third-party probe data speeds or travel times, but TrafficCube can upgrade this process by pulling in even more relevant data types, and by providing even more statistical analysis or visualizations. But while these customizable VC&V capabilities add good value, it is the automation of a rigorous data-driven calibration methodology that more directly demonstrates the kind of unique capability that TrafficCube can offer the industry.

The lack of available software automation for FHWA's TAT Volume 3 methodology has continued to inhibit industry's adoption of the methodology ever since its publication in 2019. Some researchers and developers have tried to automate TAT Volume 3 via spreadsheet-based tools. Such tools are familiar, transparent, and easy to modify for a single corridor or study. However, the spreadsheet approach becomes less practical as the number of locations, time intervals, days, measures, clusters, and model alternatives increases. TAT Volume 3 is inherently multi-dimensional: analysts must compare observed and simulated measures across space, time, days, operating conditions, and calibration iterations. In a spreadsheet, these dimensions often become distributed across many worksheets, linked formulas, manually maintained ranges, charts, and summary tables. That structure can work for a small demonstration, but it is difficult to scale, audit, reproduce, or adapt when the underlying data change. It also places a heavy burden on the analyst to preserve consistent indexing, formulas, filters, and chart definitions across repeated calibration cycles.

TrafficCube has more potential as a practical implementation platform because it treats the TAT Volume 3 workflow as a structured data and analysis problem rather than as a collection of spreadsheet calculations. By organizing traffic data into a consistent cube with location, time, and date or scenario dimensions, the tool can preserve the relationships among observed data, simulation outputs, clusters, representative days, variation envelopes, and calibration diagnostics. This makes it possible to automate repeated steps such as episode screening, cluster assignment, representative-day selection, envelope generation, and comparison of observed and simulated measures without rebuilding formulas or charts by hand. The cube structure also supports visual inspection of spatial and temporal congestion patterns, including upstream propagation and persistence across time intervals, while maintaining a

consistent analytical frame for statistical summaries. In this sense, TrafficCube does not replace the judgment required by TAT Volume 3. Instead, it provides a more scalable and repeatable environment for applying that judgment to realistic, multi-source traffic datasets.

TrafficCube can also allow agencies and practitioners to apply selected portions of the TAT Volume 3 procedure without committing to the full methodology in every study. For example, a user could apply the clustering and representative-day functions to a year of observed data, identify a small number of days that capture recurring operating conditions, and then simulate only those representative days rather than immediately proceeding through every subsequent calibration step. Similarly, once alternatives analysis is implemented, users may be able to compare simulated strategies statistically within the cube environment even when they do not have a full year of field-measured data available for formal calibration. This modularity is important because real-world studies often face schedule, budget, data availability, and staffing constraints. TrafficCube can therefore support both rigorous applications that follow the full Volume 3 workflow and more focused applications that borrow specific methods, such as clustering, envelope comparison, or alternatives testing, to improve the structure and transparency of practical traffic analysis.

As of early October 2026, TrafficCube had implemented the first three steps of the 4-step methodology (i.e., 1) ingestion, fusion, and normalization of multi-source data, 2) cluster analysis and identification of representative days, 3) variation and calibration envelope development, 4) alternatives analysis), with step 4 likely completed by the end of October. But TrafficCube may further increase the value of Volume 3 analysis through its visualization of multi-source data, its ability to perform specific sub-steps of the methodology when users want, and its embedded intelligence for interpreting results and recommending actions (see figure 6). An appropriate disclaimer is that practitioners have not yet applied TrafficCube's TAT Volume 3 analysis to a wide variety of real-world projects, which have the potential to expose implementation errors. However, the AI-enhanced (i.e., Codex) source code analysis is expected to make any such errors relatively fast and easy to rectify.

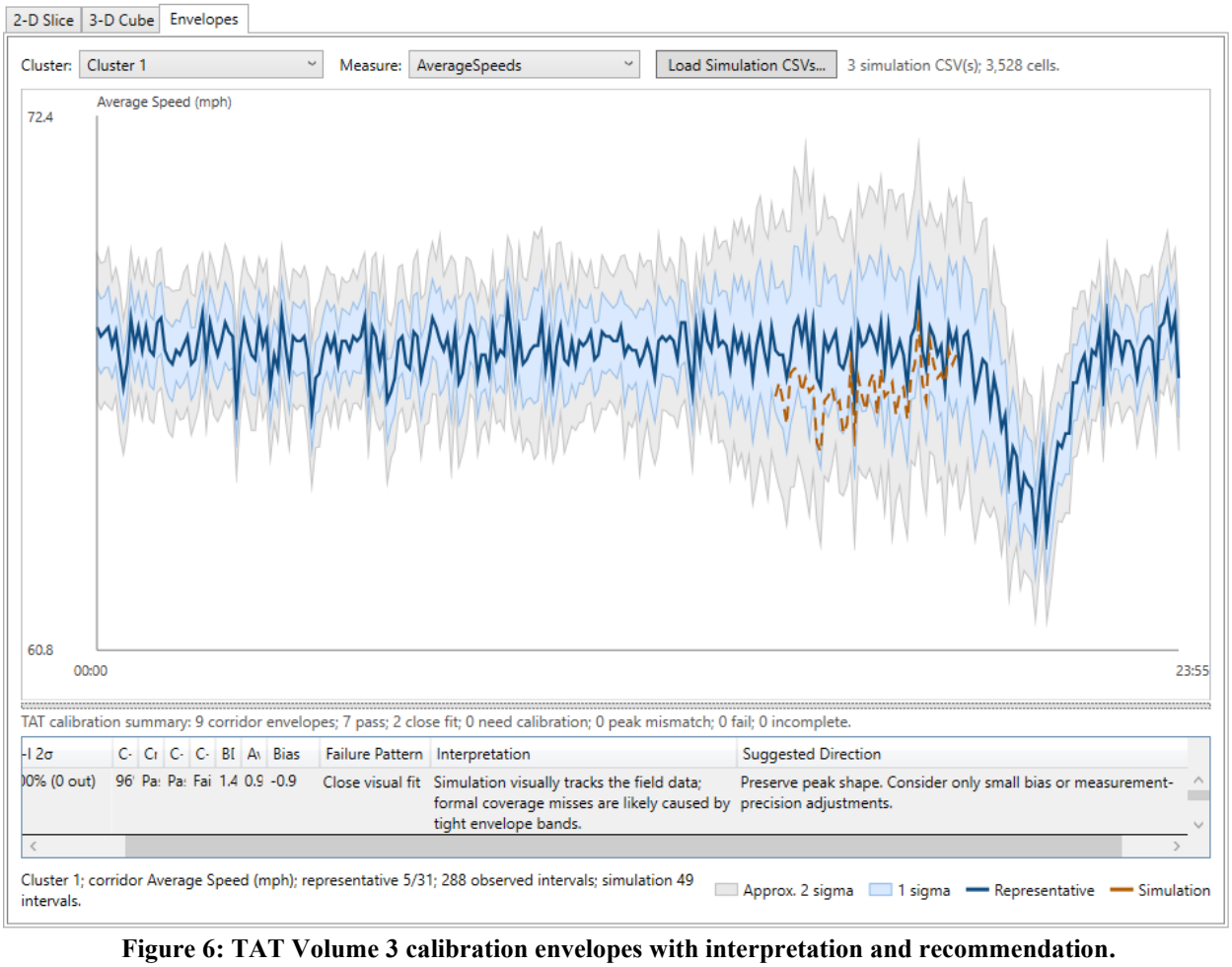


Figure 6: TAT Volume 3 calibration envelopes with interpretation and recommendation.

Multi-Day Analysis

Traffic analysts often need to examine traffic network changes across many weeks or months. TrafficCube now includes specific multi-day analysis methods and visualizations to assist in such studies. The properties panel lets the user jump to specific percentile days of the year (or percentile scenarios, when the Z-axis represents scenarios instead of dates). Next, before-and-after analyses help in quantifying and visualizing the effects of a change in the traffic network. To facilitate these studies, TrafficCube reports percentile metrics while providing line charts (see figure 7), histograms, probability density functions (see figure 8), scatterplots, and cumulative distribution functions. Finally, before-and-after analyses are also related to travel time reliability analysis studies, which do not necessarily include a before-and-after cutoff date. However, figure 7 and figure 8 are intended to illustrate the travel time reliability effects of a variable speed limit (VSL) system deployed halfway through the year. For example, by comparing the numeric and visual outputs associated with average speeds, travel times, and speed variances, an analyst could determine that 1) VSL did not affect average speeds or travel times very much, but 2) VSL significantly reduced speed variances and excessive travel time days during the second half of the year.

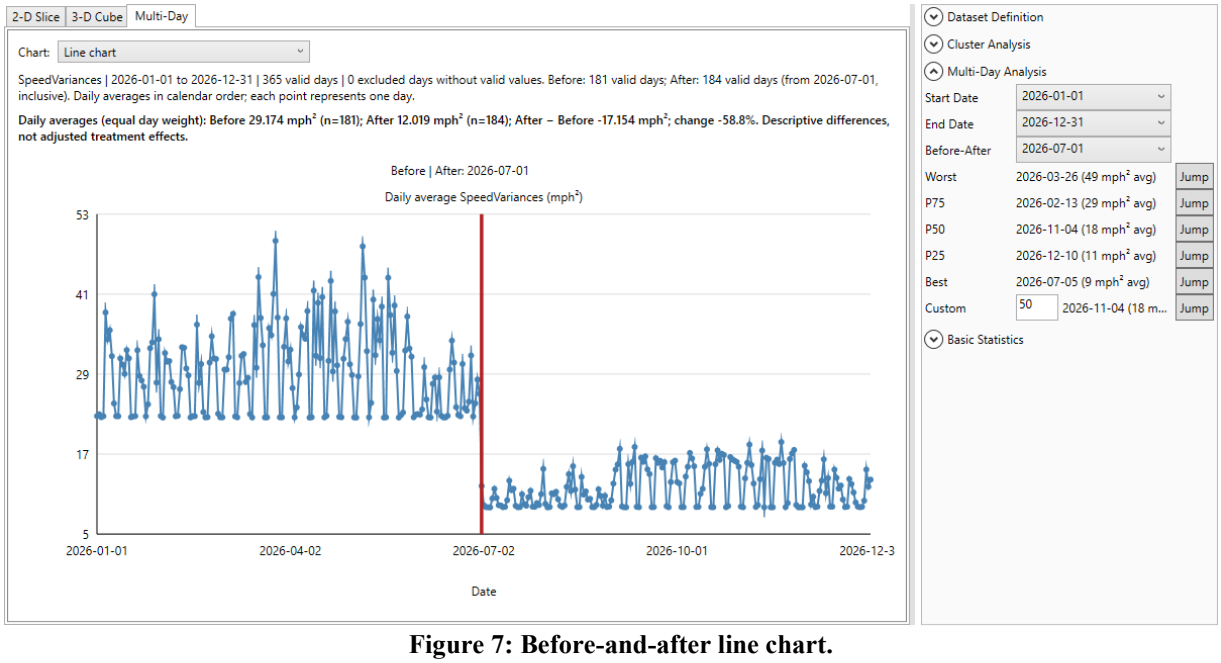


Figure 7: Before-and-after line chart.

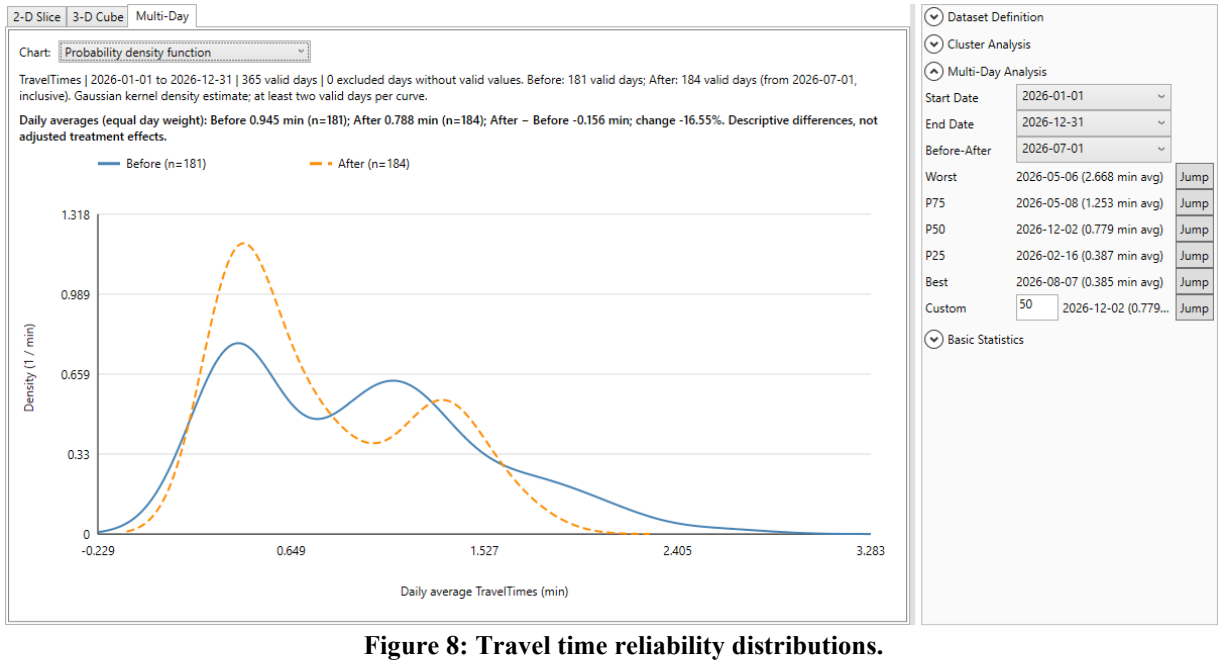


Figure 8: Travel time reliability distributions.

High-Value Planned Methodologies

The future looks bright for data analytics and AI to augment traditional traffic analysis, modeling, and simulation (AMS). The following are some planned methodologies that may allow TrafficCube to contribute to this movement.

Similarity searches

Similarity search is one promising application of the TrafficCube concept. Once traffic conditions are organized by location, time interval, date, and measure, the software can search for historical periods that resemble a current day, selected congestion episode, simulated scenario, or user-defined operating condition. For example, an analyst could select a severe bottleneck pattern and ask the tool to identify other days with similar upstream speed degradation, travel time growth, weather conditions, demand levels, or incident timing. This capability would help users move beyond single-day inspection toward pattern recognition across large archives of traffic data. Similarity search could also support representative-day selection, before-and-after comparisons, scenario matching, and short-term prediction by identifying previous conditions that developed in comparable ways.

Agentic AI applications

With TrafficCube having the pedigree to perform a wide variety of methodological calculations and analyses while operating on any desired types of transportation data, there will be opportunities for human researchers and AI agents to apply the program numerous times in succession, to answer key questions for a single project. Because of this, the program will need mechanisms for running calculations in batch mode without the UI. The program will also need multiple forms of well-styled and well-structured output reports for maximum understanding by humans and machines alike. The program will also need extensive libraries of user guide documentation for humans and AI agents, although large language models (LLMs) will certainly expedite the production of such documentation.

Congestion causal pie charts

Being able to quantify the specific sources of traffic congestion in a locality can be key towards guiding congestion mitigation investments for the State or local transportation agency. More than two decades ago, FHWA published a national version of the congestion causal pie chart, helping to clarify the typical sources of urban traffic congestion (10). Following this seminal research and proof-of-concept, multiple agencies and researchers have developed methods and tools for constructing data-driven versions of the pie chart for their local jurisdictions (11, 12). The major online platform RITIS now offers a Causes of Congestion tool capable of generating national, regional, or corridor specific pie charts informed by connected vehicle probe data and crowdsourced content (13, 14). Because TrafficCube can similarly access weeks, months, or years of multi-source data needed to construct the pie chart, the program will increase the adoption of such valuable analyses to wider audiences of practitioners and researchers. In addition, TrafficCube will bring additional multi-factor analysis capabilities not available in the other platforms. This includes the ability to statistically quantify and visualize the top causes, times, and locations of traffic incidents and accidents (e.g., crash causal pie charts and heat maps), plus optional integration with traffic simulation data.

Before-and-after assessment

The *Transportation Systems Simulation Compendium* devoted an entire chapter to post-project validation, while conceding that agencies and practitioners do not perform such validation nearly as often as they should (15). To be fair, traffic analysis projects rarely provide sufficient schedule and budget for such validations. However, such validations are potentially critical to identifying chronic AMS biases or suboptimal practices. Moreover, beyond traditional AMS, agencies simply need concrete data on the before-and-after impacts of their transportation infrastructure and active traffic management (ATM) investments. In addition, agencies need real-time travel time reliability monitoring systems, to flag system disruptions in need of immediate attention (16). These post-project validations, before-after studies, and real-time monitoring are thus additional functions and capabilities that TrafficCube can democratize and facilitate. But beyond the traditional studies or monitoring that agencies and consultants currently perform, this time it will be possible to perform more advanced statistical and AI-enhanced evaluations of traffic network or traffic corridor performance. TrafficCube could support stronger before-after evaluations of ATM systems by organizing before-and-after data into a consistent multi-dimensional structure. Instead of treating the evaluation as a small number of aggregate before-and-after comparisons, the tool could preserve the location, time-of-day, day-of-week, weather, volume, and deployment-period dimensions needed to create matched comparison periods. This structure would help analysts compare similar operating conditions before and after implementation, normalize results by volume, isolate treated segments from control segments where available, and

avoid drawing conclusions from unmatched or atypical periods. Take variable speed limit (VSL) systems as an example. The cube structure could help analysts avoid overstating safety benefits by making a corridor's pre-deployment crash and operating-history context more visible, including whether the corridor was analyzed during an unusually bad crash period that might have improved somewhat even without the VSL system. TrafficCube could make the operational performance of VSL more transparent by letting users store, visualize, and compare multiple speed metrics within the same analytical frame. Analysts could use the 85th-percentile speed to assess driver response to posted limits, mean speed to support mobility and benefit-cost calculations, and speed variance to indicate whether the system is achieving speed harmonization by narrowing the gap between faster and slower vehicles. By viewing these measures across segments, time intervals, weather conditions, and traffic volumes, analysts could distinguish between broad system effects and localized or temporary changes.

Data quality and anomaly identification

Because the tool compares measures across consistent cube dimensions, it can flag values that appear inconsistent with nearby locations, adjacent time intervals, related measures, or expected traffic behavior. Examples could include missing records, repeated flat-line values, impossible speeds, abrupt discontinuities, demand values that greatly exceed plausible capacity, travel times inconsistent with observed speeds, or isolated extreme cells that do not propagate spatially or temporally like real congestion. These checks would not eliminate the need for analyst judgment, but they could make data issues easier to find before formal analysis begins. In this role, TrafficCube would function as both an analytical tool and a data-screening environment, helping users distinguish meaningful transportation patterns from sensor errors, formatting problems, incomplete feeds, or unrealistic simulation outputs.

Machine learning and short-term prediction

TrafficCube creates opportunities for future machine learning and short-term prediction methods because it organizes transportation data into the kind of consistent, multi-dimensional structure that predictive models require. A machine learning model is only as useful as the training data made available to it. TrafficCube's ingestion process can align traffic measures, weather conditions, incidents, speed limits, crash records, and other operational variables by location, time interval, and date or scenario. This structure could support supervised learning methods that estimate near-term congestion, travel time, crash risk, speed compliance, or the likely persistence of queues after an incident or weather event. For example, a model could be trained on historical cube layers to recognize combinations of demand, weather, incident occurrence, and upstream speed changes that often precede a breakdown in traffic flow. The prediction output could then be written back into the cube as another layer, allowing users to compare observed, simulated, and predicted conditions within the same visual and statistical framework. A University of Utah VSL evaluation provides one example of this broader opportunity (17). The researchers used machine learning methods, including support vector machine and artificial neural network models, to estimate crash frequency and severity from roadway, traffic, environmental, and crash-related inputs. Future TrafficCube implementations could support similar methods by helping analysts prepare the required input features, screen for missing or imbalanced data, compare model predictions with observed outcomes, and visualize where prediction errors are concentrated across space, time, weather, or operating conditions. This would be especially useful for short-term prediction because the cube can preserve the sequence of conditions leading up to an event, rather than treating each observation as an isolated record. In this role, TrafficCube would not need to prescribe a single machine learning algorithm. Its more important contribution would be to provide a practical data structure and user interface for building, testing, interpreting, and communicating predictive transportation models. Then after demonstrating accuracy of the predictive methods on historical data, developers could quickly implement the same methods within real-time traffic management and control systems.

AI-based traffic analysis insights and recommendations

A further opportunity is to use AI methods to generate traffic analysis insights and recommendations from the structured cube data. Because TrafficCube can preserve the relationships among demand, weather, incidents, speed, travel time, travel time index, clusters, representative days, and calibration results, an AI-assisted module could help summarize likely causes of congestion, identify recurring bottlenecks, and suggest next diagnostic steps. For example, the system might explain that a recurring speed drop begins at a specific downstream segment during the PM peak, propagates upstream for several intervals, and becomes more severe when demand is high and weather or incident indicators are present. Over time, these summaries could be informed by a curated corpus of traffic analysis guidance, including HCM concepts, bottleneck diagnosis methods, calibration guidance, and agency practice. The goal would not be to replace professional judgment, but to help analysts move more efficiently from large multi-dimensional datasets to defensible interpretations, questions, and recommendations.

TrafficCube's Niche within the Industry

Instead of being known as an AI tool, TrafficCube's cleaner identity may be as a spatiotemporal analytics engine that can support AI, feed AI, validate AI outputs, be embedded into AI-enabled workflows, and in some cases implement AI via open-source libraries. TrafficCube does not seek to compete directly with major online integrated operating environments (e.g., Traffic Mobility Insights, DAVE, RITIS). Those are enterprise systems having cloud platforms, internal data governance, secure data pipelines, conversational interfaces, and production workflows. TrafficCube's more plausible industry role is as a method demonstrator that helps define what those larger systems should eventually be able to do, and as an all-purpose tool that can make data analytics more accessible and user-friendly for the general public. TrafficCube's spatiotemporal reasoning could make it valuable even if large agencies already have dashboards, geographic information systems (GIS) platforms, or AI chat interfaces. Its near-term value is as an all-purpose platform for spatiotemporal traffic analyses, especially where analysts need to examine how traffic conditions vary jointly by location, time, scenario, day, weather, incident condition, or simulation seed. The strongest future industry contribution may be in analyses that current tools often do awkwardly:

1. Make spatiotemporal analytics more concrete. Let users open, inspect, test, and critique once-opaque methods.
2. Serve as a proving ground for calculations and innovative research. If a new metric or novel methodology works well in TrafficCube, it can then go into Caltrans-like systems, vendor platforms, or agency data environments.
3. Democratize and demystify advanced analysis. Large departments of transportation may build enterprise AI platforms, but smaller agencies, researchers, and consultants still need low-cost, lightweight tools for focused studies.
4. Preserve the value of TAT Volume 3. The industry adoption barrier since 2019 has not been lack of interest, it has been lack of software automation. In this regard, TrafficCube could provide a bridge from method to practice.

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